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Evaluation of IMERG climate trends over land in the TRMM and GPM eras

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Abstract

The Integrated Multi-satellitE Retrievals for global precipitation measurement (GPM) mission (IMERG) is a global precipitation product suite consisting of both near-real-time and research-grade products with high spatiotemporal resolution. However, the IMERG developers note that it is designed as a high resolution precipitation product (HRPP), not a Climate Data Record, and its ability to capture climate trends remains uncertain. Therefore, it is imperative to explore and quantify IMERG's capability in capturing climate trends for our community. This study examines the climatological performance of IMERG Final (Version 06) by analyzing annual precipitation trends over land from 2001 to 2019 using the Mann-Kendall analysis, taking gauge records as ground truth. Three different matching strategies are applied: at gauge locations, at 0.1° pixels, and at 1.0° pixels. Additionally, this study compares the performances during the TRMM (2001–2014) and GPM eras (2015–2019). Our results find: (1) IMERG daily data exhibits high spatial consistency with gauge records at both gauge locations and 0.1° resolution, consistent with its Global Precipitation Climatology Centre calibration, with a conversion rate of approximately 89.3%; (2) IMERG performs much better overall in the GPM era than in the tropical rainfall measuring mission (TRMM) era, evidenced by a lower proportion of unreliable samples (∼10.2% vs. ∼21.2%); (3) The proportion of samples showing consistent trends with gauge data is 86.7% in the GPM era, much higher than the 70.5% and 75.3% shown by the entire record and the TRMM era, respectively. This improvement in the GPM era suggests that the within-mission consistency of IMERG is higher than the between-mission consistency, likely due to residual differences in the calibration methods used during the TRMM and GPM missions. This study broadens the perspective on IMERG, showcasing its additional potential for analyzing climate trends despite its design only as a HRPP. Crucially, it highlights and reconfirms how the GPM era has enhanced IMERG's capacity for accurately tracking global precipitation trends.

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1. Introduction

Precipitation, a fundamental component of the Earth's water cycle, is playing a pivotal role in the planet's climate dynamics (Trenberth 2011, Liu *et al* 2012, Hong and Gourley 2014). It serves as one of the critical indicators of climate change impact, exerting far-reaching influences on various biological, ecological, and hydrological processes (Kidd and Huffman 2011). The quantification and understanding of precipitation patterns are essential for the scientific community and policymakers to address issues related to climate change, water resource management, and disaster risk assessment (Wen *et al* 2018, Li *et al* 2022b, Liu and Yao 2024).

The National Aeronautics and Space Administration (NASA) global precipitation measurement (GPM) mission science team provides the Integrated Multi-satellitE Retrievals for GPM (IMERG) as a state-of-the-science precipitation product suite based on multi-source satellite observations (Huffman *et al* 2015, 2020). Currently, IMERG has been proven to be effective in real-time precipitation observation and the assessment of hydrological events with high spatiotemporal resolutions (0.1° and half hourly; (Pradhan *et al* 2022, Tang *et al* 2016, Luo *et al* 2021, Zhu *et al* 2021a). Some efforts have been coordinated to examine the validity of IMERG in climate-related studies. For example, Huffman *et al* (2023) presented the new Version 3.2 of the global precipitation climatology project, which integrates various precipitation estimates (including IMERG) for improved time and space homogeneity.

Despite the IMERG developers' statement in the technical documentation that 'IMERG is not designed to be computed as a Climate Data Record', there is still a body of work using IMERG for precipitation trend analysis (Huffman *et al* 2015). Zhang *et al* (2023) employed IMERG data from 2001 to 2020 to investigate the distributions of global precipitation systems, finding that large-size precipitation systems exhibit nonsignificant increasing trends, while small and medium-sized precipitation systems exhibit significant decreases in the study period. Ghanghas *et al* (2023) found that the increasing global temperature is correlated with a substantial contraction of precipitation coverage in tropical regions, yet an expansion of precipitation coverage in arid areas. Saouabe *et al* (2022) combined IMERG with five regional climate models over a Moroccan semi-arid basin, and showed that the decrease in precipitation (up to 47%) will lead to a decrease in surface runoff ranging up to 45%. By analyzing IMERG in the past two decades, Li *et al* (2022a) found an increasing trend in both the annual precipitation amount (up to 50 mm yr^{-1}) and frequency (up to 20 d yr^{-1}) in most regions of the Tibetan Plateau, during which light and moderate precipitation contribute approximately 70% and 90% to the increases in regional mean precipitation amount and frequency, respectively. We note that, in common with other high resolution precipitation products, the high spatial resolution of IMERG surpasses that of other precipitation products, such as Global Precipitation Climatology Centre (GPCC), which are designed for climate research. A careful assessment of its utility is therefore imperative (Ghanghas *et al* 2023, Hussein *et al* 2021, Saouabe *et al* 2022, Li *et al* 2022a, Zhang *et al* 2023). The challenge of using IMERG for climate research lies in its inhomogeneous retrievals attributed to changes in the population of constellation satellites, retrieval skill differences among sensors, and infrequent satellite overpasses, with complications due to weather regimes, topographic features, and radiometrically challenging surface types (Zhu *et al* 2018, Li *et al* 2022c, Wang *et al* 2023). For example, Li *et al* (2020) pointed out that IMERG underestimates extreme precipitation due to sensor insensitivity or saturation. Meanwhile, IMERG significantly underestimates precipitation over Tibet Plateau with total errors ∼50% for both tropical rainfall measuring mission (TRMM) and GPM eras (Tang *et al* 2018). Zhu *et al* (2021b) found strong negative correlations between IMERG metrics and elevation, latitude, and distance from the seashore, based on their evaluations over Mainland China. These uncertainties lead to discrepancies when compared to *in-situ* rain gauges (the ground truth), which may hamper climate studies.

This study aims to address: (1) how reliable is GPM IMERG in depicting climate trends over land? (2) How do the TRMM and GPM eras compare in capturing precipitation trends? We employ Mann-Kendall trend analysis to compare the temporal trends of IMERG predictions and Gauge measurements from 2001 to 2019 at three different spatial resolutions including the gauge location, 0.1° pixels, and 1.0° pixels.

2. Materials

2.1. IMERG

From 2000 to 2014, IMERG was built upon TRMM, which laid the foundation for the retrieval of precipitation from space-based platforms with the precipitation radar and TRMM microwave imager in the tropics and subtropics. Starting in March 2014, the GPM mission succeeded TRMM, incorporating dual-frequency precipitation radar (DPR) and the GPM microwave imager (GMI), which extended coverage to the mid-latitudes and sub-polar regions.

IMERG provides three distinct data runs—Early, Late, and Final—each designed to meet different demands for timeliness and accuracy (Huffman *et al* 2019). The IMERG final run daily V06 data, used in this study, includes all available data, including gauge data from GPCC, thus providing the best accuracy

for detailed climate studies and hydrological analysis (Huffman *et al* 2020). Additionally, we downloaded the corresponding IMERG V06 monthly data to assess its performance in NASA's calibration process. Both the daily and monthly datasets were directly obtained from NASA's official website, ensuring data integrity and consistency. Utilizing IMERG V06 ensures consistency with previous research that primarily used this version when it was the latest available (Ghanghas *et al* 2023, Li *et al* 2022a, Zhang *et al* 2023). Furthermore, the enhancements in Version 07 mainly improve the capture of extreme rainfall events, which have minimal impact on climate-scale analyses (Huffman *et al* 2022), but are important for the recent increasing focus on the climate of extreme events. We did an additional sensitivity test using the IMERG V07 product, which has a longer time record.

2.2. GPCC

The GPCC provides high-quality global precipitation data essential for climate monitoring and research. The GPCC collects raw data from rain gauges over the globe, applies rigorous quality control, and uses advanced interpolation techniques to produce accurate and reliable gridded datasets. This study uses the GPCC Monitoring Product at its native monthly and 1-degree resolution, serving as the reference data for calibrating IMERG monthly data (Schneider *et al* 2008, 2016, Schamm *et al* 2014). IMERG utilizes the GPCC Full Analysis for retrospective processing as far as it is available and then shifts to the Monitoring Analysis thereafter. This approach is crucial because the sampling in this combination is strongly time-dependent, with more gauges present earlier in the record and fewer later on, including some notable drops in gauge counts along the way. It is worth noting that this time-dependent sampling can significantly influence the consistency and reliability of IMERG data across different time periods, which is a key consideration when interpreting trends and patterns in the data.

2.3. Gauge records

The Serially Complete Earth (SC-Earth) dataset is a global daily dataset spanning from 1950 to 2019, which includes key meteorological variables like precipitation, mean daily temperature, temperature range, dew-point temperature, and wind speed (Tang *et al* 2021). Although the data are not strictly aligned with UTC 00:00–24:00, the annual averaging process will largely mitigate these biases. The dataset is derived from global historical climatology network-daily and Global Surface Summary of the Day sources after multiple merging strategies and strict quality control. In SC-Earth, there are a total of 64 399 global gauges with daily records of precipitation, which are regarded as ground truth to represent precipitation. In this study, we employ SC-Earth data as the reference dataset to evaluate the performance of IMERG in identifying the spatial-temporal precipitation trends. Note that the GPCC Monitoring Product is generated based on approximately 7000 stations, whereas SC-Earth dataset in this study includes about 64 000 stations.

3. Methodology

3.1. Spatial matching of satellite and gauge precipitation data

In this study, aligning ground-based gauge stations with satellite-derived IMERG pixel data is the starting point for validating satellite precipitation estimates. We consider points at gauge locations, 0.1° pixels, and 1.0° pixels as the basic units for matching IMERG samples with gauge records. For gauge location, each gauge was matched to one pixel, totaling 64 399. For the 0.1° grid, the total number of gauge-occupied grid boxes is 50 722. For the 1.0° grid, the total number of gauge-occupied grid boxes is 8164. The alignment process begins with the calculation of matrix indices (row index for latitude and column index for longitude) that relate the latitude/longitude coordinates of gauge stations to the IMERG spatial grid. For the matching strategy based on gauge locations, we extract the precipitation values for the nearest IMERG grid box center that matches each station's location. This enables a direct comparison between gauge-recorded precipitation and satellite-derived precipitation, with each gauge record corresponding to one sample in that pixel. For the matching strategies based on 0.1° and 1.0° pixels, if multiple gauges are located at the same pixel, their values are averaged. Based on these matching strategies, we generated 64 399 samples at the gauge locations, 50 722 samples at 0.1° pixels, and 8184 samples at 1.0° pixels. Subsequently, we matched the temporal resolution of IMERG data with gauge measurements by accumulating precipitation data annually to analyze yearly changes. The chosen method for averaging is the arithmetic mean, treating the missing value as nan records, and regarding 0 mm h^{-1} as no precipitation.

3.2. Mann-Kendall trend analysis

The Mann-Kendall trend analysis (hereafter MK) is a non-parametric statistical tool used to detect monotonic trends in time series data (Hamed and Rao 1998). The MK test provides a robust way to determine the significance of trends, particularly in the presence of outliers or non-normal data, making it a valuable tool for analyzing time series data. Compared to linear regression analysis, this method is less influenced by the magnitude of trends and extreme values, placing greater emphasis on the trends themselves. In MK analysis, two key outcomes are the statistic value (referred to as S), which indicates the strength of trend

(i.e. the percentage increase or decrease of precipitation amount overall the past 20 yr), and the *P*-value (referred to as *P*), which reflects the significance level (i.e. whether the increasing and decreasing trends are statistically significant). *S* and *P* could be defined in equations (1)–(3):

$$S = \sum_{i=1}^{n-1} \sum_{j=i+1}^{n} \operatorname{sign}\left(X_j - X_i\right) \tag{1}$$

$$\operatorname{Var}(S) = \frac{1}{18}\left(n(n-1)(2n+5) - \sum_{i=1}^{n} t_i (t_i - 1)(2t_i + 5)\right) \tag{2}$$

$$P = 2 * \left(1 - \operatorname{normcdf}\left(\frac{S}{\operatorname{Var}(S)^{0.5}}\right)\right) \tag{3}$$

where *S* and *P* are statistic value and significant value, respectively; *X* is the data series, and the subscript *i* and *j* are data indexes; *n* denotes the total length number of series data; Var (*S*) is the variance of *S*; t_i represents the number of observations in the *i*th group; and normcdf calculates the cumulative distribution function of a normal distribution.

Since the *S* value from the MK trend analysis increases with the expansion of the time series length, using normalized *S* provides a more suitable measure for evaluating trend magnitudes across different time periods (equation (4)). In this process, if the *S* value is greater than zero, it is reduced by 1, and if the *S* value is less than zero, it should be increased by 1. Additionally, it is worth to note that the MK analysis in this study is conducted using IMERG daily data,

$$\text{Normalized } S = \begin{cases} (S-1)/\sqrt{\operatorname{Var}(S)} & if\, S > 0 \\ 0 & if\, S = 0 \\ (S+1)/\sqrt{\operatorname{Var}(S)} & if\, S < 0 \end{cases}. \tag{4}$$

4. Results

4.1. Verification and comparison of IMERG trends

The numerical results from the MK analysis, as shown in figure 1, alongside corresponding linear regression results, illustrate IMERG's capability to capture precipitation trends on a climatic scale. Figures 1(a)–(e) demonstrate that IMERG daily data exhibits an acceptable correlation value in linear regression at both gauge locations and a 0.1° resolution (0.51 for both) because the vast majority of 0.1° grid boxes with gauges are single-gauge. It is comparable to the correlation coefficients obtained between IMERG monthly data and gauge records (0.49 and 0.50, respectively). This indicates that IMERG's temporal downscaling strategy is effective, allowing the daily data to nearly inherit the trend-capturing ability of the monthly data. Figures 1(d)–(f) further reveal that the spatial downscaling strategy nearly preserves IMERG monthly data's trend-capturing ability, with correlation coefficients of 0.49, 0.50, and 0.51 for gauge locations, 0.1°, and 1.0° resolutions, respectively. Although GPCC data achieves the highest correlation coefficient (0.56) at a 1.0° resolution, IMERG monthly data maintains a correlation coefficient of 0.51, and the IMERG daily data achieves an acceptable correlation coefficient of 0.50 at gauge locations, resulting in a conversion rate of approximately 89.3%. The process of calibrating IMERG monthly data with GPCC is known to improve this performance. This suggests that if IMERG was calibrated with data demonstrating superior climatology capturing capabilities, its ability to capture climate trends could be substantially enhanced.

4.2. Spatial distributions of trends

The effectiveness of IMERG in capturing climate trends is quantified using the MK *S* statistic, as depicted in figure 2, where red and blue indicate increasing and decreasing trends, respectively, throughout the study period from 2001 to 2019. To assess IMERG's performance at various resolutions, we analyzed the MK *S* at gauge locations, 0.1°, and 1.0° resolutions. This analysis also considers IMERG monthly data and GPCC data to demonstrate the calibration of daily IMERG data. According to figures 2(a)–(e), daily IMERG data exhibit spatial patterns similar to gauge records, with significant increasing trends in regions like eastern Asia and western North America, and decreasing trends predominating in eastern South America, southern Africa, and central Asia. Notably, discrepancies in trends for southeastern Asia (shown in the box), with gauges indicating an increasing trend and IMERG showing a decreasing trend, are attributed to the calibration of IMERG using the GPCC. Meanwhile, the IMERG monthly data shows the high consistencies among the results from different matching strategies including gauge locations, 0.1°, and 1.0° resolutions, which is also highly consistent with the gauge records with the corresponding resolutions. The GPCC gauge-based climatology data, which features lower resolutions (monthly, 1°), helps calibrate the total monthly precipitation at a 1.0° resolution, subsequently anchoring the daily data scaling at 0.1° resolution to ensure consistency with the monthly data. Figures 2(d)–(i) demonstrate that the spatial distribution of trends is well-preserved in the resampling process from 1.0° to 0.1°, indicating that resolution is not a significant factor in trend analysis. Overall, daily IMERG data's strong consistency with gauge records in terms of spatial distribution can be attributed to the calibration process using GPCC, with monthly IMERG data serving as a vital intermediary. Additionally, IMERG's ability to capture climate trends is significantly influenced by geographic features (e.g. latitude, elevation and the distance from seashore), suggesting potential for further algorithm

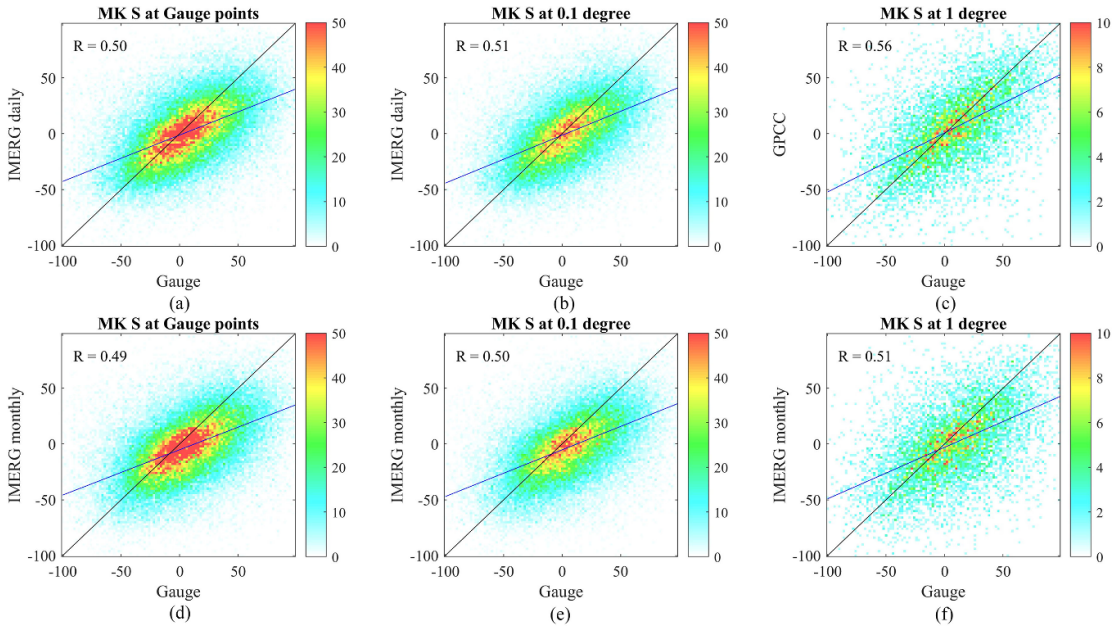


Figure 1. Numerical distribution and linear regression results of the MK *S* at different resolutions, including (a) and (b) the results between the IMERG daily data and gauge records, (c) the result between the GPCC monthly monitoring product and gauge records, and (d)–(f) the results between IMERG monthly data and the gauge records.

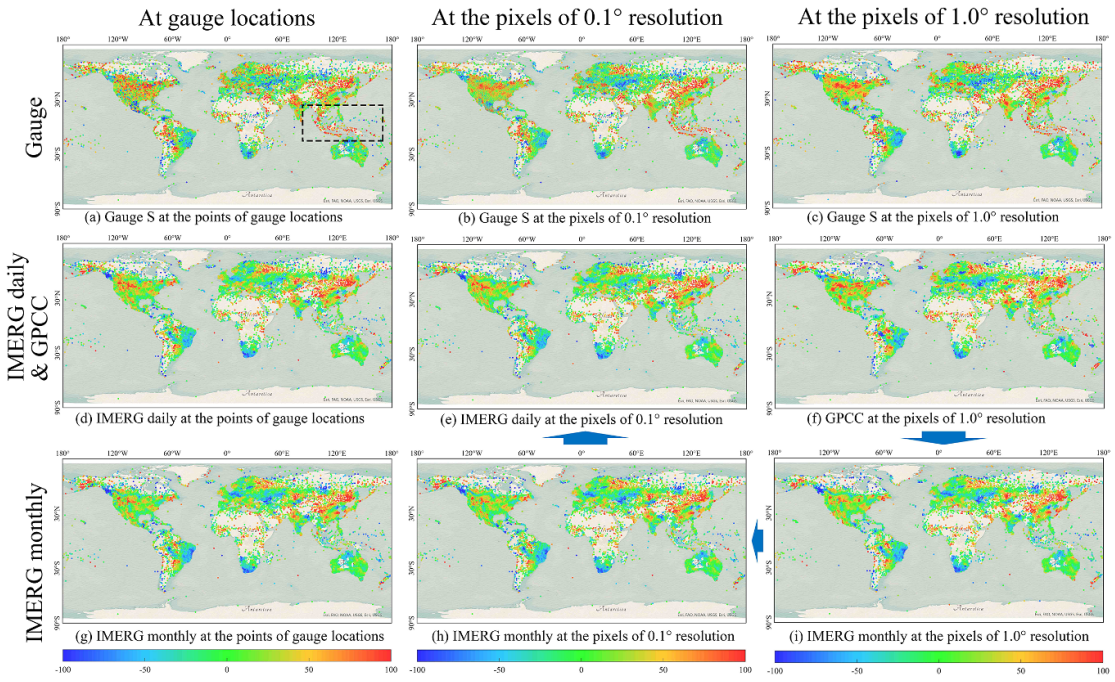


Figure 2. Spatial distribution of the MK *S*, including the (a)–(c) *S* of gauge, (d)–(e) *S* of IMERG daily, (f) *S* of GPCC, and (g)–(i) *S* of IMERG monthly at different resolutions which contains gauge locations, 0.1° and 1.0°. The arrows here indicate the sequence in which IMERG was calibrated.

refinement to improve accuracy (for details, please refer to section 2 of the supporting information)

4.3. How reliable are IMERG trends?

Since IMERG performs well at different resolutions, we also evaluated its performance across different eras, including the entire period (2001–2019), the TRMM era (2001–2013), and the GPM era (2015–2019). Note that the length of the period may influence the results of MK analysis, and thus the analysis in a shorter period will be more easily influenced by the random errors. We also did a comprehensive evaluation by taking the Climate Prediction Center dataset as the true values based on the IMERG V07 daily data from 2015 to 2023 (Please see section 3 of supplementary). The results show that the trends in

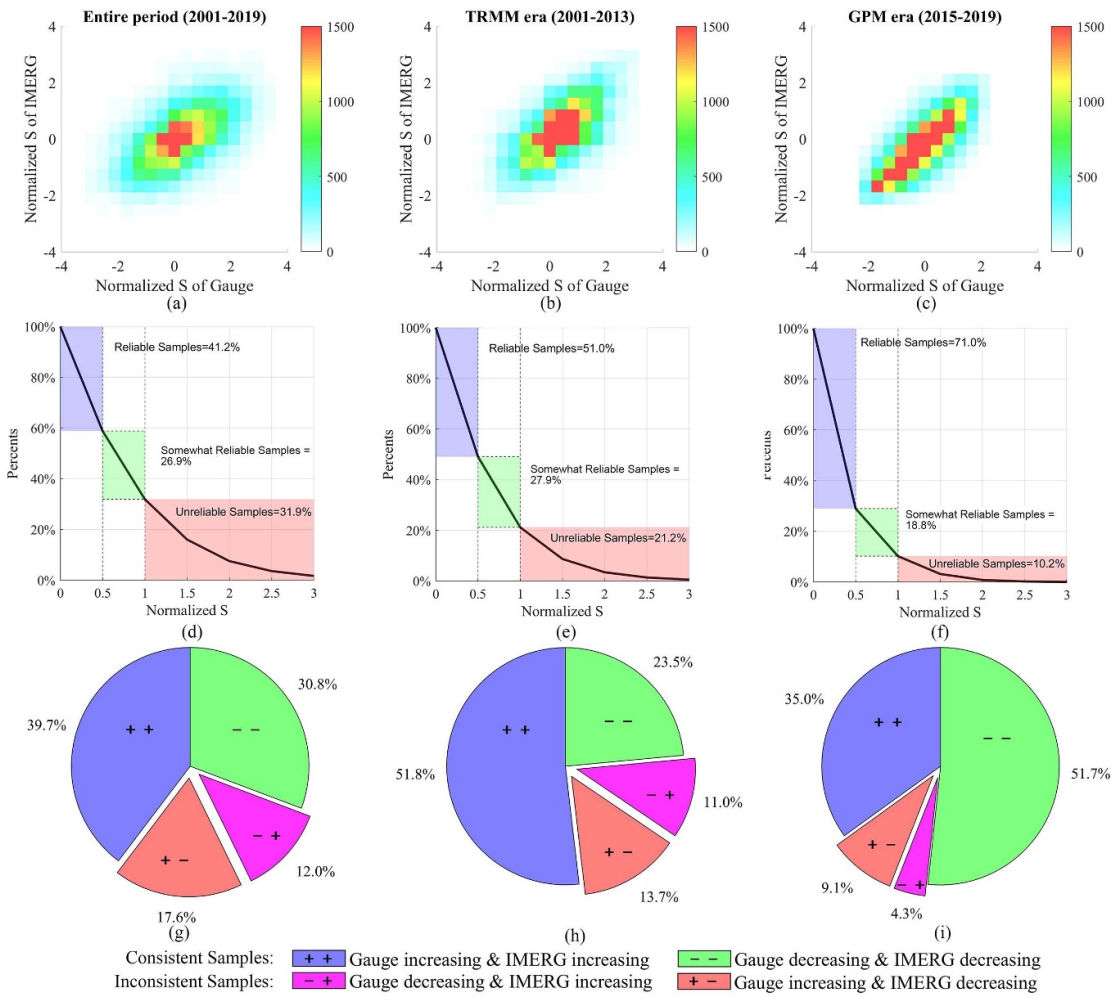


Figure 3. Statistical results of the MK analysis based on IMERG V06 daily data, including density maps of normalized *S* (a)–(c), curves showing percentages on the *y*-axis versus differences in normalized *S* on the *x*-axis (d)–(f), and pie charts illustrating the consistency of trends between IMERG and gauges (g)–(i).

GPM ear are better than TRMM era, which is consistent with this analysis.

The years 2000 and 2014 were excluded from the respective eras because IMERG in 2000 is incomplete and 2014 is the transitional year between the TRMM and GPM eras. In this study, we define 'samples' as the matched pairs between gauge records and corresponding IMERG values, and we use samples at the 0.1° resolution for temporal analysis since this is the native IMERG resolution. From the density maps of normalized *S* (figures 3(a)–(c)), IMERG shows significantly better performance in the GPM era compared to the TRMM era and the entire period, indicated by a more concentrated distribution of samples along the 1:1 line ($y = x$). This suggests that the differences in *S* values between IMERG and gauge data are smaller in the GPM era. Consequently, the samples are categorized into three groups: (1) reliable samples, with differences in normalized *S* less than 0.5; (2) somewhat reliable samples, with differences in normalized *S* ranging from 0.5 to 1; and (3) unreliable samples, with differences in normalized *S* greater than 1. These categories correspond to different segments of the percentage curves in figures 3(d)–(f). The proportion of unreliable samples in the GPM era is significantly lower (∼10.2%) compared to the entire period (∼31.9%) and the TRMM era (∼21.2%). Pie charts (figures 3(g)–(i)) also reveal that the consistency of MK *S* values between gauge and IMERG is better in the GPM era than in the entire period and the TRMM era. Specifically, the proportion of samples showing consistent trends with the gauge data (both increasing or decreasing) is 86.7% in the GPM era, compared to 70.5% and 75.3% in the entire period and the TRMM era, respectively.

Based on the classification strategy for the samples, IMERG data in the GPM era strongly outperforms that in the TRMM era (figure 4). First, it is necessary to have an overall view of the differences between the MK *S* values of IMERG and those of the gauges. From figure 4(a), it is evident that IMERG daily data calibrated by GPCC underestimates the increasing trends by a wide margin in

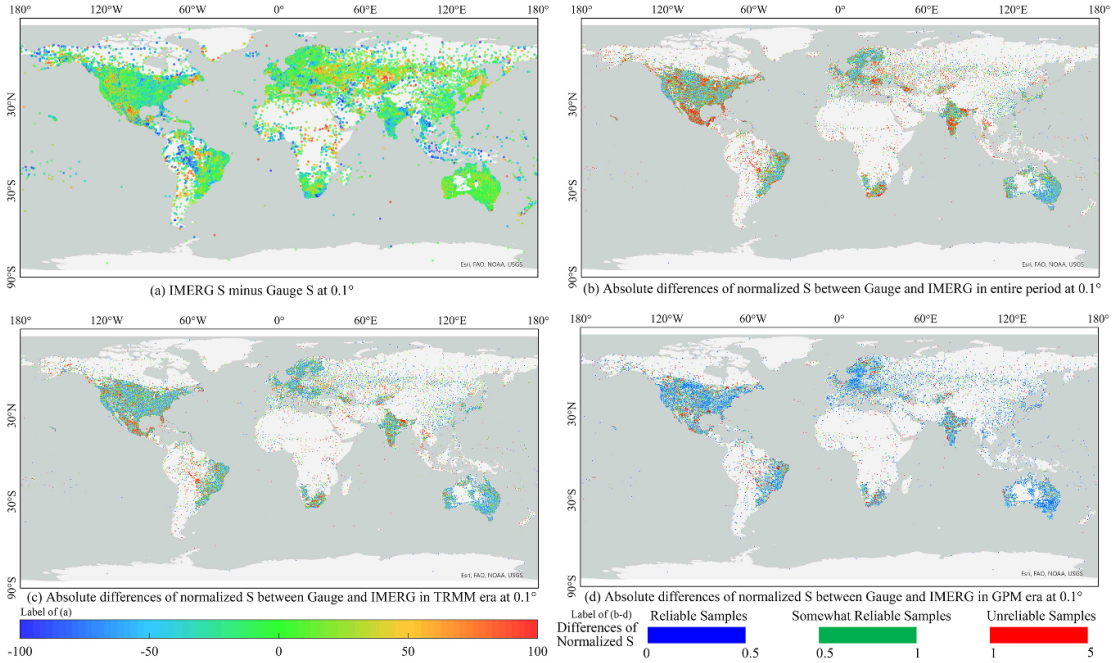


Figure 4. The spatial distributions of samples based on IMERG V06 daily data showing (a) the differences of *S* between IMERG and gauge, (b)–(d) the absolute difference of normalized *S* in entire period, TRMM era and GPM era, where the samples are divided in to reliable (blue), somewhat reliable (green) and unreliable (red) samples.

Southeast Asia and central South America. In these areas, gauges indicate strong increasing trends, while IMERG daily data shows stable or even decreasing trends. Additionally, IMERG tends to overestimate the increasing trends in a small region of central Asia. Despite these discrepancies, IMERG still demonstrates high consistency with gauge records across most regions. In figures 4(b)–(d), blue, green, and red represent reliable, somewhat reliable, and unreliable samples, respectively, with varying hues indicating different values within each category. Throughout the entire period, unreliable samples are predominantly located in Mexico, central South America, and southern India. The spatial distribution of unreliable samples during the TRMM era closely mirrors that of the entire period. In contrast, IMERG data from the GPM era demonstrates good consistency with gauge readings, with reliable and somewhat reliable samples dominating most regions. GPM-era unreliable samples are confined to a few areas such as central South America and Mexico. Notably, the classification of samples is based on their ability to capture trends, yet the results reveal coherent spatial patterns. This indicates a correlation between these characteristics and geographical and climatic features. For example, in the high latitudes of the Northern Hemisphere (e.g. central Siberia), IMERG shows overall improvement from the TRMM to the GPM era, which could be partly attributed to the absence of the GPM combined radar-radiometer algorithm (CORRA) data at higher latitudes during the TRMM era.

5. Discussion

5.1. Potential explanation of the improvements of GPM era

IMERG integrates multi-sourced satellite observational data, employing infrared (IR) observations from geostationary satellites to assist filling large gaps in passive microwave (PMW) data from low-earth-orbit satellites during the TRMM era. Although PMW data are more accurate, both data types rely on statistical relationships between observed brightness temperature and precipitation intensity, which can introduce significant systematic errors due to diverse climatic and geographical features. This issue is particularly severe with IR data, as it only captures information from the cloud top, making it less effective for shallow and stratiform precipitation (Huffman *et al* 2010, Ma *et al* 2016).

In the TRMM era, high latitude corrections in IMERG are achieved by using climatological TMI-CORRA calibrations that, extend from the edge of the TRMM PR's coverage at 35° N/S to higher latitudes. This extension is performed by utilizing GPM-era GMI-CORRA background climatological histograms and correction curves (Huffman *et al* 2015). In the GPM era, the GPM Core Observatory's 65° inclined orbit covers these higher-latitude regions that TRMM's 35° inclination cannot. As well, compared to TRMM PR, the GPM DPR supports a more physically-based precipitation retrieval algorithm and offers precise satellite-based anchor points for calibration and validation. Additionally, the scarce gauge

stations in these regions often hamper the accuracy of precipitation estimates due to the lack of sufficient ground-truth data for satellite observation calibration (Behrangi 2020, Mahmoud *et al* 2021, Milani and Kidd 2023). Furthermore, the calibration based on GPCC has also contributed much to the improvements of the precipitation estimation over the high-latitude region. Considering the significant impact of gauge reduction on trend detection, it is possible that IMERG's ability to accurately capture trends is influenced by the decrease in the gauge population of GPCC.

5.2. Potential reasons for the trend deviation between IMERG and gauges

The underlying causes of trend deviation can primarily be attributed to the following three factors: uneven gauge density, inconsistent gauge quality, and spatial variability in climatic characteristics. At the global scale, precipitation gauges are distributed unevenly, with their placement greatly influenced by population density, levels of socio-economic development, and the complexity of terrain and topography (Kidd *et al* 2017). In this study, the gauge dataset (SC-Earth) used for evaluating IMERG includes over 60 000 gauges, while the GPCC dataset, which serves to calibrate IMERG, contains only around 7000 gauges. This substantial difference in gauge coverage is a key factor contributing to the inconsistent performance between SC-Earth and IMERG. Moreover, the quality of these gauges varies significantly, since they are maintained by different countries and organizations with diverse standards and equipment. Additionally, regional climate variability contributes to trend deviations. For example, in Maritime Southeast Asia, gauge data indicates a significant increasing trend in precipitation, while IMERG data reflects a more stable trend. This region experiences a tropical climate and consists of numerous islands, and combined with global warming, which leads to frequent and intense extreme precipitation events compared to other areas. Extreme precipitation events are inherently challenging to capture accurately, contributing to uncertainty in climate trend assessments.

6. Conclusion

Given the growing interest in utilizing IMERG for global precipitation trend analyses, our study seeks to examine its utility: IMERG is capable of capturing the climate trends in land regions where gauge validation data are available and performs better in the GPM era.

The detailed findings include: (1) IMERG daily data shows high spatial consistency with the gauge records at both gauge locations and 0.1° resolutions based on the MK analysis; (2) the ability to capture the climate trends of IMERG could be attributed to the calibration process based on GPCC with the conversion rate of approximately 89.3%; (3) IMERG in GPM era significantly performs better than IMERG in TRMM era with the evidence that the proportion of unreliable samples in the GPM era is significantly lower (∼10.2%) compared to the entire period (∼31.9%) and the TRMM era (∼21.2%); and (4) the proportion of samples showing consistent trends with the gauge data (both increasing or both decreasing) is 86.7% in the GPM era, compared to 70.5% and 75.3% in the entire period and the TRMM era, respectively.

In this study, while IMERG's ability to capture climate trends effectively and GPM's role in enhancing IMERG's efficacy in tracking climatic precipitation trends is affirmed, it is important to note that IMERG is not specifically designed for climate analysis. Thus, GPCC and other climate-specific datasets are preferred.

Data Access Statement

The IMERG daily Version 06 Final Run data can be download from the NASA official website at https://gpm.nasa.gov/data/imerg; the DOI is https://doi.10.5067/GPM/IMERGDF/DAY/06. The SC-Earth data can be downloaded from the link https://zenodo.org/record/4762586; the DOI as https://doi.10.5281/zenodo.4762586. The GPCC Monitoring Analysis data can be downloaded from the GPCC website at www.psl.noaa.gov/data/gridded/data.gpcc.html; the DOI is https://doi.10.5676/DWD_GPCC/FD_M_V7_100.

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Conflict of interest

The authors declare no conflict of interest.

Ethics statements

This study contains no human subjects, human data or tissue, or animals.

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References

Behrangi A 2020 Improving high-latitude and cold region precipitation analysis *Satellite Precipitation Measurement* vol 2 pp 881–95

Ghanghas A, Sharma A, Dey S and Merwade V 2023 How is spatial homogeneity in precipitation extremes changing globally? *Geophys. Res. Lett.* **50** e2023GL103233

Hamed K H and Rao A R 1998 A modified Mann-Kendall trend test for autocorrelated data *J. Hydrol.* **204** 182–96

Hong Y and Gourley J J 2014 *Radar Hydrology: Principles, Models, and Applications* (CRC Press)

Huffman G J *et al* 2015 NASA global precipitation measurement (GPM) integrated multi-satellite retrievals for GPM (IMERG) *Algorithm Theoretical Basis Document Version* vol 4 p 30

Huffman G J *et al* 2020 Integrated multi-satellite retrievals for the global precipitation measurement (GPM) mission (IMERG) *Satellite Precipitation Measurement* (Springer) pp 343–53

Huffman G J, Adler R F, Behrangi A, Bolvin D T, Nelkin E J, Gu G and Ehsani M R 2023 The new version 3.2 global precipitation climatology project (GPCP) monthly and daily precipitation products *J. Clim.* **36** 7635–55

Huffman G J, Adler R F, Bolvin D T and Nelkin E J 2010 The TRMM multi-satellite precipitation analysis (TMPA) *Satellite Rainfall Applications for Surface Hydrology* (Springer) pp 3–22

Huffman G J, Stocker E F, Bolvin D T, Nelkin E J and Tan J 2019 *GPM IMERG Final Precipitation L3 1 day 0.1 degree x 0.1 degree V06* (Goddard Earth Sciences Data and Information Services Center (GES DISC)) (https://doi.org/10.5067/GPM/IMERGDF/DAY/06)

Huffman G, Bolvin D, Nelkin E and Tan J 2022 On the verge of IMERG version 07 Authorea Preprints

Hussein K A, Alsumaiti T S, Ghebreyesus D T, Sharif H O and Abdalati W 2021 High-resolution spatiotemporal trend analysis of precipitation using satellite-based products over the United Arab Emirates *Water* **13** 2376

Kidd C, Becker A, Huffman G J, Muller C L, Joe P, Skofronick-Jackson G and Kirschbaum D B 2017 So, how much of the Earth's surface is covered by rain gauges? *Bull. Am. Meteorol. Soc.* **98** 69–78

Kidd C and Huffman G 2011 Global precipitation measurement *Meteorol. Appl.* **18** 334–53

Li D, Qi Y and Chen D 2022a Changes in rain and snow over the Tibetan Plateau based on IMERG and Ground-based observation *J. Hydrol.* **606** 127400

Li Z, Chen M, Gao S, Hong Z, Tang G, Wen Y, Gourley J J and Hong Y 2020 Cross-examination of similarity, difference and deficiency of gauge, radar and satellite precipitation measuring uncertainties for extreme events using conventional metrics and multiplicative triple collocation *Remote Sens.* **12** 1258

Li Z, Gao S, Chen M, Gourley J J, Liu C, Prein A F and Hong Y 2022b The conterminous United States are projected to become more prone to flash floods in a high-end emissions scenario *Commun. Earth Environ.* **3** 86

Li Z, Tang G, Kirstetter P, Gao S, Li J-L F, Wen Y and Hong Y 2022c Evaluation of GPM IMERG and its constellations in extreme events over the conterminous united states *J. Hydrol.* **606** 127357

Liu Z, Ostrenga D, Teng W and Kempler S 2012 Tropical rainfall measuring mission (TRMM) precipitation data and services for research and applications *Bull. Am. Meteorol. Soc.* **93** 1317–25

Liu Z and Yao T 2024 Strategizing earth science data development *Sci. Data* **11** 693

Luo Z, Shao Q, Wan W, Li H, Chen X, Zhu S and Ding X 2021 A new method for assessing satellite-based hydrological data products using water budget closure *J. Hydrol.* **594** 125927

Ma Y, Tang G, Long D, Yong B, Zhong L, Wan W and Hong Y 2016 Similarity and error intercomparison of the GPM and its predecessor-TRMM multisatellite precipitation analysis using the best available hourly gauge network over the Tibetan Plateau *Remote Sens.* **8** 569

Mahmoud M T, Mohammed S A, Hamouda M A, Dal Maso M and Mohamed M M 2021 Performance of the imerg precipitation products over high-latitudes region of Finland *Remote Sens.* **13** 2073

Milani L and Kidd C 2023 The state of precipitation measurements at mid-to-high latitudes *Atmosphere* **14** 1677

Pradhan R K, Markonis Y, Vargas Godoy M R, Villalba-Pradas A, Andreadis K M, Nikolopoulos E I, Papalexiou S M, Rahim A, Tapiador F J and Hanel M 2022 Review of GPM IMERG performance: a global perspective *Remote Sens. Environ.* **268** 112754

Saouabe T, Naceur K A, El Khalki E M, Hadri A and Saidi M E 2022 GPM-IMERG product: a new way to assess the climate change impact on water resources in a Moroccan semi-arid basin *J. Water Clim. Change* **13** 2559–76

Schamm K, Ziese M, Becker A, Finger P, Meyer-Christoffer A, Schneider U, Schröder M and Stender P 2014 Global gridded precipitation over land: a description of the new GPCC First Guess Daily product *Earth Syst. Sci. Data* **6** 49–60

Schneider U *et al* 2016 GPCC full data reanalysis version 7.0: monthly land-surface precipitation from rain gauges built on GTS based and historic data

Schneider U, Fuchs T, Meyer-Christoffer A and Rudolf B, 2008 Global precipitation analysis products of the GPCC *Global Precipitation Climatology* vol 112 (Internet Publikation)

Tang G, Clark M P and Papalexiou S M 2021 SC-earth: a station-based serially complete earth dataset from 1950 to 2019 *J. Clim.* **34** 6493–511

Tang G, Long D, Hong Y, Gao J and Wan W 2018 Documentation of multifactorial relationships between precipitation and topography of the Tibetan Plateau using spaceborne precipitation radars *Remote Sens. Environ.* **208** 82–96

Tang G, Ma Y, Long D, Zhong L and Hong Y 2016 Evaluation of GPM Day-1 IMERG and TMPA Version-7 legacy products over Mainland China at multiple spatiotemporal scales *J. Hydrol.* **533** 152–67

Trenberth K E 2011 Changes in precipitation with climate change *Clim. Res.* **47** 123–38

Wang T, Li Z, Ma Z, Gao Z and Tang G 2023 Diverging identifications of extreme precipitation events from satellite observations and reanalysis products: a global perspective based on an object-tracking method *Remote Sens. Environ.* **288** 113490

Wen Y, Behrangi A, Chen H and Lambrigtsen B 2018 How well were the early 2017 California atmospheric river precipitation events captured by satellite products and ground-based radars? *Q. J. R. Meteorol. Soc.* **144** 344–59

Zhang Y, Li R and Wang K 2023 Climatology and changes in internal intensity distributions of global precipitation systems over 2001–2020 based on IMERG *J. Hydrol.* **620** 129386

Zhu S *et al* 2021a A morphology-based adaptively spatio-temporal merging algorithm for optimally combining multisource gridded precipitation products with various resolutions *IEEE Trans. Geosci. Remote Sens.* **60** 1–21

Zhu S, Shen Y and Ma Z 2021b A new perspective for charactering the spatio-temporal patterns of the error in GPM IMERG over mainland China *Earth Space Sci.* **8** 1–16

Zhu S, Wan W, Xie H, Liu B, Li H and Hong Y 2018 An efficient and effective approach for georeferencing AVHRR and GaoFen-1 imageries using inland water bodies *IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens.* **11** 2491–500